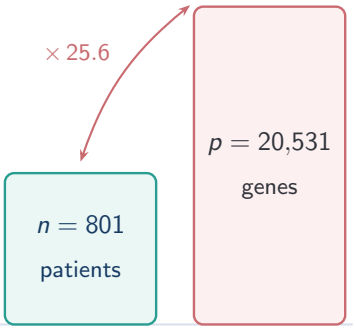
Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks on TCGA RNA-Seq Data ($p \gg n$)

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Why Is High-Dimensional Gene Expression Hard?



Statistical bottlenecks in $p \gg n$:

- **Singular covariance:** $\mathrm{rank}(\mathbf{S}) \leq 800 \ll 20{,}531$; classical LDA fails.
- **Severe multicollinearity:** co-expressed genes ($|r| > 0.85$) destabilize selection.
- **High-variance estimation:** unconstrained models overfit noise without shrinkage.

▷ **Regularization is essential for well-posed estimation.**

SECTION 01

Dataset and Exploratory Analysis

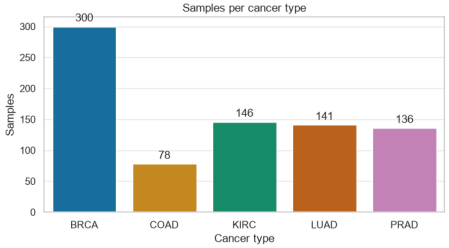
TCGA RNA-Seq Dataset

801 samples
Total patients

5 classes
BRCA KIRC COAD
LUAD PRAD

1,000 genes
MAD pre-screen

20,531 genes
Raw transcriptome

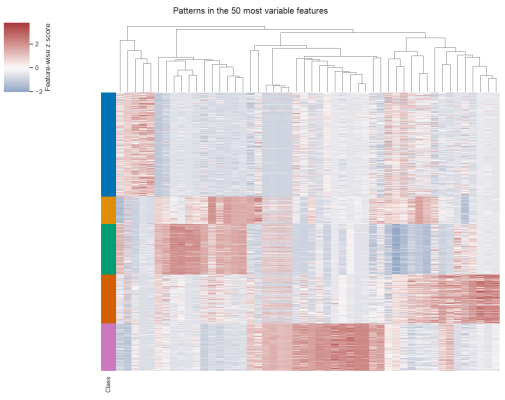


Preprocessing pipeline:

- **Log-transform:** $z_{ij} = \log_2(x_{ij} + 1)$ stabilizes expression variance.
- **Feature screening:** Top-1,000 genes by in-fold MAD (zero test leakage).
- **Standardization:** In-fold Z-score scaling ($\mu = 0$, $\sigma = 1$) per feature.

Expression Heatmap: 50 Highest-Variance Genes

Unsupervised hierarchical clustering reveals tissue-specific expression blocks



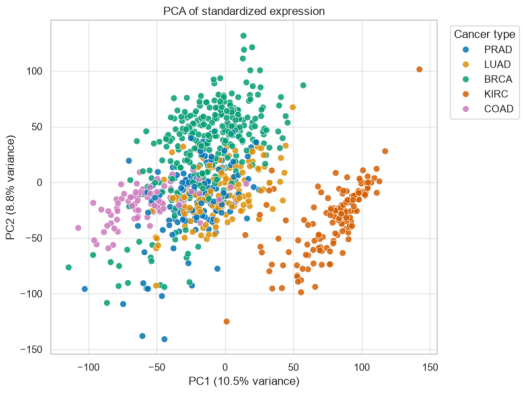
Key observations:

- Colours show standardised expression per gene; columns are patients.
- Cancer cohorts form **distinct blocks** without any labels.
- Row clusters reflect shared **gene modules** and pathways.
- Strong visual separation motivates linear classification.

▷ **The dataset carries strong class-specific structure even in unsupervised clustering.**

PCA: Tumours Separate in Two Linear Dimensions

Spectral projection of 1,000 variance-filtered genes onto the leading principal components



- **PC1 (18.6%):** Separates BRCA from KIRC.
- **PC2 (12.3%):** Separates COAD and LUAD.
- **30.9%** of total variation visible in 2D.
- Five cohorts are almost fully linearly separable in this unsupervised projection.

▷ **Linear separability exists before any model fitting — non-linear methods gain little here.**

SECTION 02

Classification Models

Multinomial Logistic Regression & Regularization

Model class probabilities using the softmax function:

$$P(Y_i = k \mid x_i) \;=\; \frac{\exp(x_i^\top \beta_k)}{\sum_{j=1}^{5} \exp(x_i^\top \beta_j)}$$

Estimate coefficients by minimizing multiclass cross-entropy with a regularization penalty:

$$\mathcal{L}(B) = \mathcal{L}_{\text{CE}}(B) + \lambda P(B)$$

3. Regularization strategies

Model	Penalty	Statistical effect
Ridge	$\frac{1}{2}\|B\|_F^2$	Shrinks coefficients
Lasso	$\|B\|_1$	Produces sparse coefficients
Elastic Net	$\alpha\|B\|_1 + \frac{1-\alpha}{2}\|B\|_F^2$	Combines shrinkage and sparsity

Non-Linear Classifiers

1. Random Forest

- Ensemble of bootstrap-trained trees.
- Random feature subsets at each split.
- Captures nonlinearities and feature interactions.

2. XGBoost

- Gradient-boosted decision trees.
- Sequentially corrects previous prediction errors.
- Captures complex nonlinear decision boundaries.

3. PCA-MLP

- PCA reduces the dimensionality of gene expression.
- Multilayer perceptron learns nonlinear mappings.
- Combines dimensionality reduction with neural modelling.

4. Sparse Group-Lasso NN

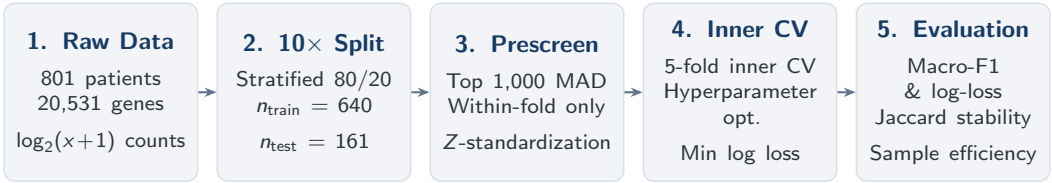
- Neural network with group-based sparsity.
- Encourages selective use of input features.
- Combines nonlinear learning with feature regularization.

SECTION 03

Experimental Design

Zero-Leakage Evaluation Pipeline

10 repeated stratified splits with strictly within-fold feature prescreening



▷ **Strict zero-leakage: feature screening and standardization are isolated within training folds.**

Why Three Metrics?

Accuracy, probability calibration, and biomarker stability

Macro-F1 (Accuracy)

$$\frac{1}{5}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k+R_k}$$

- Harmonic mean across classes.
- **Ceiling:** all models > 0.99.

Log-Loss (Calibration)

$$-\frac{1}{N}\sum_{i,k} y_{ik}\log\hat{p}_{ik}$$

- Penalizes overconfidence.
- **Decisive metric** for ranking.

Stability (Biomarkers)

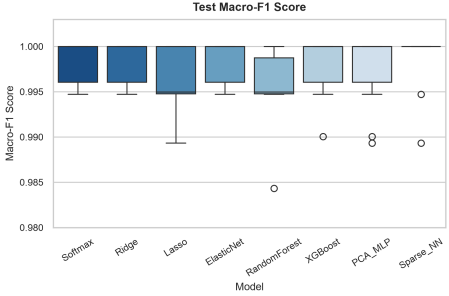
$$J = \frac{|S_a \cap S_b|}{|S_a \cup S_b|}$$

- Gene overlap across splits.
- **Key** for clinical biomarkers.

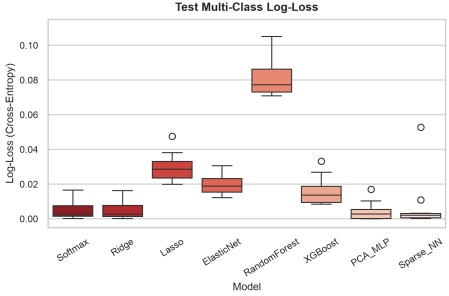
SECTION 04

Results

All Models Achieve High Accuracy — Log-Loss Tells the Real Story (Performance distributions across 10 repeated train/test splits)



F1 saturates above 0.993 for all models; accuracy alone cannot distinguish architectures.



Log-loss diverges: tree models (0.081) vs. regularized linear models (< 0.020).

Comprehensive Model Comparison

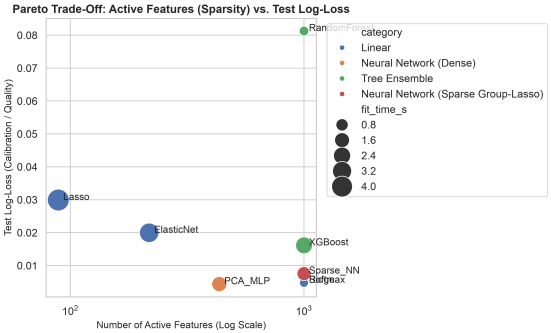
10-split cross-validated test results

Model	Macro-F1	Log-Loss	Genes	Jaccard
Softmax	0.998	0.005	1000	—
Ridge	0.998	0.005	1000	—
Lasso	0.996	0.030	**89**	0.740
Elastic Net	**0.998**	**0.020**	**218**	**0.773**
Random Forest	0.995	0.081	1000	0.242
XGBoost	0.998	0.016	1000	0.467
PCA-MLP	0.997	0.004	434	—
Sparse NN	0.998	0.008	1000	—

▷ **Elastic Net matches top F1 with 218 genes and leads sparse stability.**

Sparsity vs. Calibration: The Pareto Frontier

Elastic Net achieves optimal balance between features and calibration

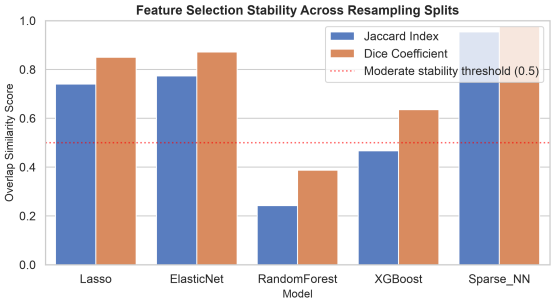


- **Ridge / PCA-MLP:** Dense (1000 genes / 434 PCs), log-loss 0.004–0.005.
- **Lasso:** Highly sparse (89 genes), log-loss 0.030.
- **Elastic Net:** Optimal trade-off (218 genes, log-loss 0.020).
- **Tree models:** Off frontier (log-loss 0.081).

▷ **Elastic Net occupies the optimal sparse Pareto frontier.**

Feature Selection Stability

Selection consistency across 10 independent data splits

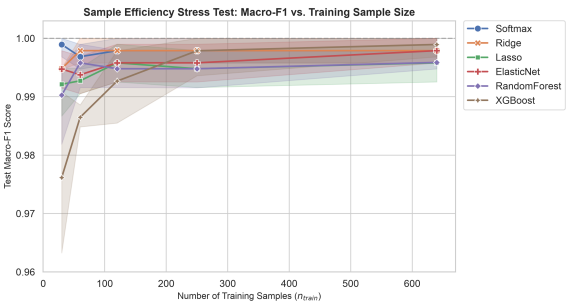


- **Elastic Net:** Jaccard 0.77 (Dice 0.87) — leads sparse models.
- **Lasso:** Jaccard 0.74 — collinear omission hurts stability.
- **XGBoost:** Jaccard 0.47 — greedy splits fluctuate.
- **Random Forest:** Jaccard **0.24** — lowest consistency.

▷ L_2 **grouping is essential for reproducible biomarker discovery.**

Sample Efficiency Under Data Scarcity

Macro-F1 as training size drops from 640 to 30



- **Ridge & Elastic Net:** Maintain $F_1 \geq 0.995$ down to $n_{\text{train}} = 30$.
- **XGBoost:** Drops to 0.976 under data scarcity.
- **Random Forest:** Degrades to 0.985 at $n = 30$.

Y-axis starts at 0.96 to highlight low-n divergence.

▷ **Regularized linear models remain sample-efficient; trees degrade sharply.**

SECTION 05

Interpretation and Conclusions

Three Main Findings

Answering the three primary research questions of this study

Q1 **Flexibility provides no advantage in RNA-Seq tumour classification.**

Linear separability holds in gene space. Complex non-linear models yield zero F1 gain and worse calibration.

Q2 **High collinearity severely destabilises Lasso and decision trees.**

Lasso selects arbitrarily among correlated genes; tree stability drops to 0.24. Elastic Net grouping recovers stability (0.77).

Q3 **Elastic Net achieves the optimal sparsity–stability–calibration trade-off.**

Ties top Macro-F1 (0.998), eliminates 78% of candidate genes (218 retained), and delivers highest sparse biomarker consistency.

Limitations

1. Single-Cohort Evaluation

Analysis uses 801 samples from one dataset; generalizability to independent cohorts is unverified.

2. High-Dimensional Feature Space

Screening 20,531 genes to 1,000 potentially excludes relevant biological features.

3. Limited Biological Interpretability

Anonymized gene identifiers restrict biological interpretation of selected features.

4. Broad Cancer Classification

Distinguishes five cancer types without examining molecular subtypes within each type.

Thank You